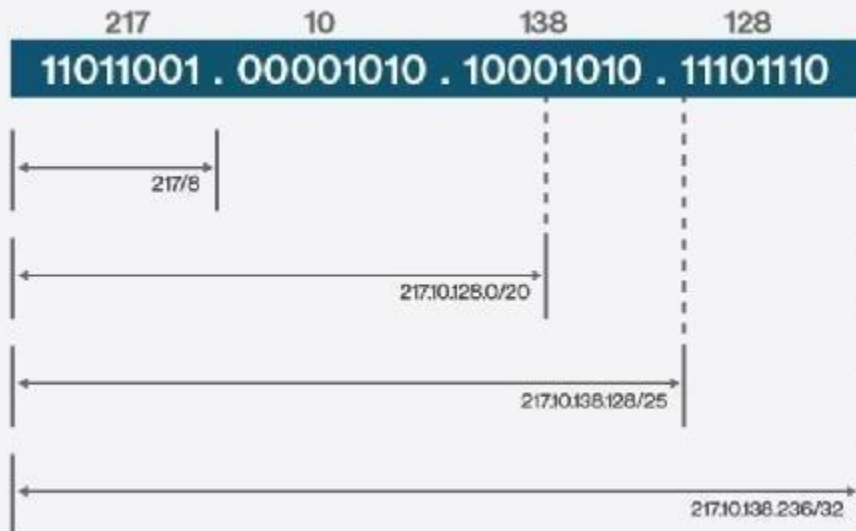
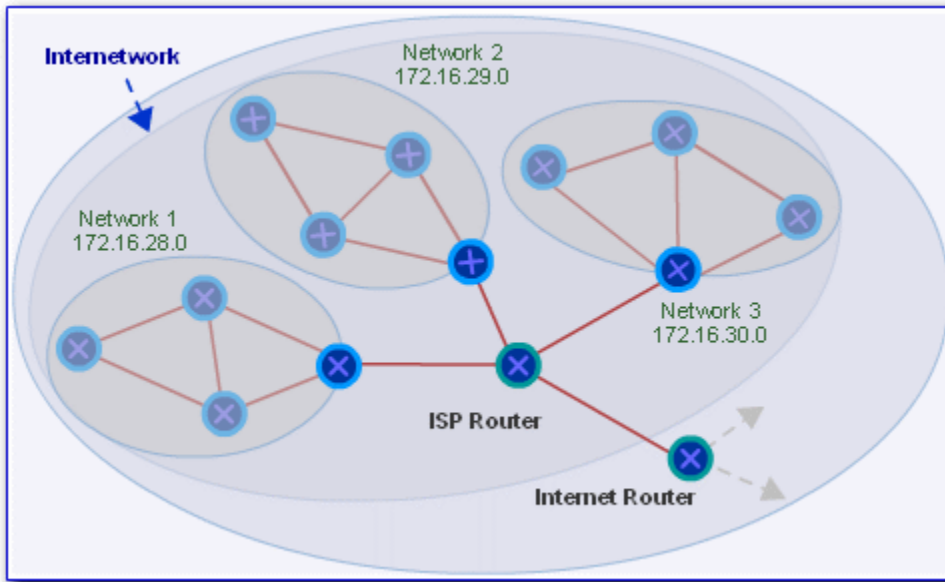


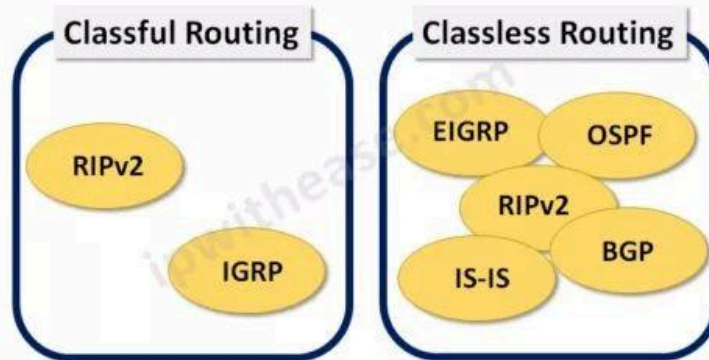
Technical Briefing: Supernetting in Classful and Classless Addressing

A precise, rule-driven synthesis with classful vs. classless intuition



Source: Technology Training Ltd.

Classful Routing vs Classless Routing



networkinterview.com
(An Initiative By ipwithease.com)

CLASSFUL ADDRESSING VERSUS CLASSLESS ADDRESSING

CLASSFUL ADDRESSING	CLASSLESS ADDRESSING
An IP address allocation method that allocates IP addresses according to five major classes	An IP address allocation method that is designed to replace classful addressing to minimize the rapid exhaustion of IP addresses
Less practical and useful	More practical and useful
Network ID and host ID changes depending on the classes	There is no boundary on network ID and host ID
	Visit www.PEDIAA.com

1. What Supernetting Really Is (and Is Not)

Supernetting—also called *route aggregation* or *summarization*—is the process of **combining two or more networks into a single CIDR prefix**.

It is the **inverse of subnetting**:

- **Subnetting:**
One large network → many smaller networks
- **Supernetting:**
Many small networks → one larger network

Core Purpose

- Reduce routing table size
- Improve router lookup speed
- Enable scalable inter-domain routing (CIDR)



Non-negotiable rule:

Supernetting is meaningless for a *single* network.

Aggregation requires **at least two networks**.

2. The Three Mandatory Rules of Supernetting

If **any one** of these rules fails, supernetting is **invalid**.

1 Contiguity Rule

All networks must be **numerically adjacent**.

- No gaps
- No overlaps
- IP ranges must form one continuous interval

Classful shortcut intuition

- **Class C:** check sequential third-octet values (e.g., 32, 33, 34, 35)
- **Class B:** still focus on the third octet, since default network ends at 16 bits

2 Uniform Size + Power of Two

Two separate but related requirements:

- All networks must be the **same size** (same prefix)
- Number of networks must be **2ⁿ** (2, 4, 8, 16...)

Valid

Invalid

2 × /24 3 × /24

4 × /24 5 × /24

8 × /26 mixed /24 + /26

③ Divisibility Rule (Most Critical)

The **first network ID** must be divisible by the **total size of the supernet**.

Binary Interpretation

If total size = 2^n addresses, then:

- The **last n bits** of the first network ID must be **all zeros**

This guarantees:

- Proper alignment
- A single CIDR prefix can represent the entire block



Exam mindset:

Most aggregation failures occur due to divisibility—not contiguity.

3. Classful Addressing Perspective

Although supernetting is inherently **classless**, exams and legacy explanations often frame it in classful terms.

Default Class Masks

Classes	Network Bits	Host Bits	Default Mask
A	8	24	255.0.0.0
B	16	16	255.255.0.0
C	24	8	255.255.255.0

Key Classful Insight

Supernetting **borrow bits from the Network ID**, so:

- The supernet mask has **fewer 1s** than the default class mask
- This is the *opposite* of subnetting

4. Subnet Mask vs Supernet Mask (Critical Comparison)

Feature	Subnetting	Supernetting
Bit Borrowing	From Host ID	From Network ID
Mask 1s	$\geq$ default class bits	$<$ default class bits
Applies to	Single network	≥ 2 networks
Result	Smaller networks	Larger network
Goal	Reduce IP waste	Reduce routing entries

Mask Recognition Rule (High-Value Exam Trick)

For a **Class C** address:

- ≥ 24 ones $\rightarrow$ **Subnet mask**
- < 24 ones $\rightarrow$ **Supernet mask**

Example:

- **255.255.252.0** $\rightarrow$ 22 ones
 $\rightarrow$ **Supernet mask for Class C**

5. Range Analysis & Redundancy

When examining routing tables:

Overlapping Networks

If one prefix **fully contains** another:

- The smaller network is **redundant**
- It can be ignored for aggregation

Example:

- 164.0 – 167.255
covers
- 166.0 – 167.255

Only the **larger prefix** matters.

6. Supernet Calculation Workflow

Step-by-Step Logic

1. **Confirm contiguity**
2. **Verify same prefix**
3. **Check power-of-two count**
4. **Apply divisibility rule**
5. **Derive supernet prefix**

Worked Example: Four Class C Networks

Given

- 4 × /24 networks
- Each = 256 addresses

Total size

- $4 \times 256 = 1024 = 2^{10}$

Host bits

- 10

New prefix

- $32 - 10 = /22$

Mask

- 255.255.252.0

Supernet ID

- First network's ID

✓ Valid aggregation **only if divisibility holds**

7. Mathematical Identity to Remember

Subnetting:

Prefix ↑ (more 1s)

Supernetting:

Prefix ↓ (fewer 1s)

This single identity prevents most conceptual errors.

